

ADDITIONAL PROBLEM SHEET 4

Topics covered: Chapters 10–11.

1. Turing conditions (Exam section A type)

Consider the following *non-dimensionalised* reaction–diffusion equation for the scalar densities u and v :

$$\frac{\partial u}{\partial t} = \nabla^2 u + \gamma F(u, v), \quad (1a)$$

$$\frac{\partial v}{\partial t} = D \nabla^2 v + \gamma G(u, v), \quad (1b)$$

where $D > 0$ and $\gamma > 0$ are constants. We assume *no-flux* boundary conditions on a Cartesian domain $[0, L_1] \times [0, L_2]$ with coordinates (x_1, x_2) . The Turing conditions for pattern formation are

$$\begin{aligned} F_u + G_v &< 0, & F_u G_v - G_u F_v &> 0, \\ G_v + D F_u &> 0, & (G_v + D F_u)^2 - 4D(F_u G_v - G_u F_v) &> 0, \end{aligned} \quad (2)$$

where we have used the notation

$$\begin{aligned} F_u &= \left. \frac{\partial F}{\partial u} \right|_{u=u_0, v=v_0}, & F_v &= \left. \frac{\partial F}{\partial v} \right|_{u=u_0, v=v_0}, \\ G_u &= \left. \frac{\partial G}{\partial u} \right|_{u=u_0, v=v_0}, & G_v &= \left. \frac{\partial G}{\partial v} \right|_{u=u_0, v=v_0}. \end{aligned}$$

- State the conditions for a homogeneous equilibrium of eq. (1).
- Define what is meant by a pattern in this context, give the explicit mathematical form for the pattern and describe what the conditions in eq. (2) enforce.
- State a condition on D under the assumption that $F_u > 0$.

2. Periodic boundary conditions (Exam section B type)

Consider the following generic reaction–diffusion system,

$$\frac{\partial u}{\partial t} = \nabla^2 u + \gamma F(u, v), \quad (3a)$$

$$\frac{\partial v}{\partial t} = D \nabla^2 v + \gamma G(u, v), \quad (3b)$$

with D and γ positive constants. We assume a Cartesian domain $[0, L_1] \times [0, L_2]$ with coordinates (x_1, x_2) . We assume the densities u and v have periodic boundary conditions

$$\begin{aligned} u(0, x_2) &= u(L_1, x_2), & u(x_1, 0) &= u(x_1, L_2), \\ v(0, x_2) &= v(L_1, x_2), & v(x_1, 0) &= v(x_1, L_2). \end{aligned} \quad (4)$$

- (a) State what is meant by a homogeneous equilibrium and state the equations of such equilibria in this case.
- (b) Consider an expansion $u(x, t) = u_0 + \varepsilon u_1(x, t)$ and $v(x, t) = v_0 + \varepsilon v_1(x, t)$, show that, to $\mathcal{O}(\varepsilon)$, solutions to u_1 and v_1 take the form

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} e^{\lambda(i,j)t} \left[C_1^{ij} \sin\left(\frac{i\pi}{L_1}x_1\right) \sin\left(\frac{j\pi}{L_2}x_2\right) + C_2^{ij} \sin\left(\frac{i\pi}{L_1}x_1\right) \cos\left(\frac{j2\pi}{L_2}x_2\right) \right. \\ \left. + C_3^{ij} \cos\left(\frac{i2\pi}{L_1}x_1\right) \sin\left(\frac{j\pi}{L_2}x_2\right) + C_4^{ij} \cos\left(\frac{i2\pi}{L_1}x_1\right) \cos\left(\frac{j2\pi}{L_2}x_2\right) \right], \quad (5)$$

where the C_α^{ij} are constants. You must state the explicit form of the functions $\lambda(i, j)$.

- (c) What would the linearised solution look like if we had instead chosen Dirichlet boundary conditions, i.e.

$$\begin{aligned} u(x_1, 0) &= u(x_1, L_2) = u(0, x_2) = u(L_1, x_2) = u_0, \\ v(x_1, 0) &= v(x_1, L_2) = v(0, x_2) = v(L_1, x_2) = v_0, \end{aligned} \quad (6)$$

where u_0 and v_0 are the homogeneous equilibrium values?

- (d) We have now considered two distinct sets of boundary conditions. In which case are the Turing conditions

$$F_u + G_v < 0, \quad F_u G_v + F_v G_u > 0, \quad (7)$$

necessary for pattern formation? Justify your answer with respect to the linear solutions for u_1 in both cases.

3. Nonlinear diffusion (Exam section B type)

Consider the following variant of a reaction–diffusion system,

$$\frac{\partial u}{\partial t} = \nabla^2 u + F(u, v) + \nabla u \cdot \nabla(v^2), \quad (8a)$$

$$\frac{\partial v}{\partial t} = D\nabla^2 v + G(u, v) + v\nabla^2 u, \quad (8b)$$

with $D > 0$ constant. We assume a Cartesian domain $[0, L_1] \times [0, L_2]$ with coordinates (x_1, x_2) , and we also assume the densities u and v satisfy no-flux boundary conditions.

- (a) State the equations of homogeneous equilibria of this system
- (b) State explicitly the boundary conditions of the system.
- (c) Show that the Turing conditions required for pattern formation take the form

$$\begin{aligned} F_u + G_v &< 0, \quad F_u G_v - F_v G_u > 0, \\ DF_u + G_v - v_0 F_v &> 0, \quad (DF_u + G_v - v_0 F_v)^2 - 4D(F_u G_v - F_v G_u) > 0. \end{aligned} \quad (9)$$

- (d) Consider the case

$$F(u, v) = u(1 - u) - uv, \quad G(u, v) = v(1 - v) + \gamma uv, \quad (10)$$

where γ is a real constant. Find all the equilibria for which at least one of u or v is 0.

- (e) For each of the equilibria found in the previous question, establish whether the Turing conditions can be satisfied. If they can be satisfied state the domain of γ values for which patterns can form.
- (f) *Good practice bonus question:* Find the other possible nonzero equilibrium and state conditions on γ (if any) for which permit pattern formation. I am happy to look at/discuss your solutions if you bring them to me.

4. Patterns and polar coordinates (Exam section A, 2017)

Consider the following *non-dimensionalised* reaction–diffusion equation for the scalar densities u and v

$$\frac{\partial u}{\partial t} = \nabla^2 u + \gamma F(u, v), \quad (11a)$$

$$\frac{\partial v}{\partial t} = D \nabla^2 v + \gamma G(u, v), \quad (11b)$$

where $D > 0$ and $\gamma > 0$ are constants. We assume *no-flux* boundary conditions on a Cartesian domain $[0, L_1] \times [0, L_2]$ with coordinates (x_1, x_2) . The Turing conditions for pattern formation are

$$\begin{aligned} F_u + G_v &< 0, & F_u G_v - G_u F_v &> 0, \\ G_v + D F_u &> 0, & (G_v + D F_u)^2 - 4D(F_u G_v - G_u F_v) &> 0, \end{aligned} \quad (12)$$

where we have used the notation

$$\begin{aligned} F_u &= \left. \frac{\partial F}{\partial u} \right|_{u=u_0, v=v_0}, & F_v &= \left. \frac{\partial F}{\partial v} \right|_{u=u_0, v=v_0}, \\ G_u &= \left. \frac{\partial G}{\partial u} \right|_{u=u_0, v=v_0}, & G_v &= \left. \frac{\partial G}{\partial v} \right|_{u=u_0, v=v_0}, \end{aligned}$$

with (u_0, v_0) a homogeneous equilibrium of the system.

- (a) State the conditions for a homogeneous equilibrium of eq. (11).
- (b) Define what is meant by a pattern in this context, give the explicit mathematical form for the pattern and describe what the conditions in eq. (12) enforce.
- (c) What would be the mathematical form of the patterns if the domain had instead been an annulus, covered with a polar coordinate system (r, θ) ? You do *not* need to state the condition on the pattern numbers k_s which results from the boundary conditions.

5. More Turing analysis (Exam section B, 2017)

Consider the following generic reaction–diffusion system

$$\begin{aligned} \frac{\partial u}{\partial t} &= \nabla^2 u + \gamma F(u, v) + \nabla^2 uv, \\ \frac{\partial v}{\partial t} &= D \nabla^2 v + \gamma G(u, v), \end{aligned}$$

where D is the ratio of diffusion of v and u and γ is a positive constant. We assume that nothing can leave or enter the system and that the domain V is Cartesian.

- (a) State the boundary conditions of this system.
- (b) State the equations which would determine the homogeneous equilibria (u_0, v_0) of this system.
- (c) Perform the steps of the Turing analysis for this system to show that there will be a growing inhomogeneous pattern for the system when

$$\begin{aligned}
F_u + G_v &< 0, \quad F_u G_v - F_v G_u > 0, \\
G_v(1 + v_0) + DF_u - G_u u_0 &> 0, \\
[G_v(1 + v_0) + DF_u - G_u u_0]^2 - 4D(1 + v_0)(F_u G_v - F_v G_u) &\geq 0,
\end{aligned}$$

where we have used the notation

$$\begin{aligned}
F_u &= \left. \frac{\partial F}{\partial u} \right|_{u=u_0, v=v_0}, \quad F_v = \left. \frac{\partial F}{\partial v} \right|_{u=u_0, v=v_0} \\
G_u &= \left. \frac{\partial G}{\partial u} \right|_{u=u_0, v=v_0}, \quad G_v = \left. \frac{\partial G}{\partial v} \right|_{u=u_0, v=v_0}.
\end{aligned}$$

- (d) Now we set

$$F = u \left(\frac{k_1 v}{k_2 + v} - u \right), \quad G = (u - 1)(v - 1), \quad (13)$$

where the constants k_1 and k_2 are positive. Find the three equilibria of the system and in each case state whether they can satisfy the inequalities

$$\begin{aligned}
F_u + G_v &< 0, \quad F_u G_v - F_v G_u > 0, \\
G_v(1 + v_0) + DF_u - G_u u_0 &> 0.
\end{aligned}$$

6. Boundary condition effects (Exam section A type)

Consider the following *non-dimensionalised* reaction–diffusion equation for the scalar densities $u(x, t)$ and $v(x, t)$,

$$\begin{aligned}
\frac{\partial u}{\partial t} &= \nabla^2 u + \gamma F(u, v), \\
\frac{\partial v}{\partial t} &= D \nabla^2 v + \gamma G(u, v),
\end{aligned}$$

where $D > 0$ and $\gamma > 0$ are constants. We assume *no-flux* boundary conditions on a Cartesian domain $[0, L_1] \times [0, L_2]$ with coordinates (x_1, x_2) . The Turing conditions for pattern formation are

$$\begin{aligned}
F_u + G_v &< 0, \quad F_u G_v - G_u F_v > 0, \\
G_v + DF_u &> 0, \quad (G_v + DF_u)^2 - 4D(F_u G_v - G_u F_v) > 0,
\end{aligned} \quad (14)$$

where we have used the notation

$$\begin{aligned}
F_u &= \left. \frac{\partial F}{\partial u} \right|_{u=u_0, v=v_0}, \quad F_v = \left. \frac{\partial F}{\partial v} \right|_{u=u_0, v=v_0} \\
G_u &= \left. \frac{\partial G}{\partial u} \right|_{u=u_0, v=v_0}, \quad G_v = \left. \frac{\partial G}{\partial v} \right|_{u=u_0, v=v_0},
\end{aligned}$$

with (u_0, v_0) a homogeneous equilibrium of the system.

- (a) Define what is meant by a pattern in this context, give the explicit mathematical form for the pattern and describe what the conditions in eq. (14) enforce.
- (b) Consider a system which is identical to the one considered in part (a), except with differing boundary conditions. Consider the following two (distinct) possibilities:
 - (i) Homogeneous periodic boundary conditions along both directions.
 - (ii) Fixed (homogeneous) values $u = u_0$ and $v = v_0$ on the $x_1 = 0, x_1 = L$ boundaries (Dirichlet boundary conditions) and no-flux boundary conditions on the $x_2 = 0, x_2 = L$ boundaries (for both u and v).

Argue in each case if the Turing conditions,

$$F_u + G_v < 0, \quad F_u G_v - G_u F_v > 0, \quad (15)$$

are necessary for pattern formation as defined in part (a).

7. Enzyme-driven Turing patterns (Exam section A type)

Consider the following reaction–diffusion model of hair growth in the *Acetabularia* sea plant. The hair growth hormone u and calcium concentration v which interact to stimulate tip growth is represented by the following set of partial differential equations:

$$\frac{\partial u}{\partial t} = \nabla^2 u + \gamma(a - u + u^2 v), \quad (16a)$$

$$\frac{\partial v}{\partial t} = D \nabla^2 v + \gamma(b - u^2 v). \quad (16b)$$

Here D, γ, a and b are positive constants and we impose no-flux boundary conditions. It is assumed spontaneous tip growth occurs due to a Turing instability of the system. We consider this system on an annular domain, with no-flux boundary conditions imposed.

- (a) The homogeneous equilibria of the system are:

$$u_0 = a + b, \quad v_0 = u_0 - a. \quad (17)$$

The Turing conditions for pattern formation are

$$\begin{aligned} F_u + G_v &< 0, & F_u G_v - G_u F_v &> 0, \\ G_v + D F_u &> 0, \end{aligned} \quad (18)$$

and

$$(G_v + D F_u)^2 - 4D(F_u G_v - G_u F_v) > 0,$$

where we have used the notation

$$\begin{aligned} F_u &= \left. \frac{\partial F}{\partial u} \right|_{u=u_0, v=v_0}, & F_v &= \left. \frac{\partial F}{\partial v} \right|_{u=u_0, v=v_0}, \\ G_u &= \left. \frac{\partial G}{\partial u} \right|_{u=u_0, v=v_0}, & G_v &= \left. \frac{\partial G}{\partial v} \right|_{u=u_0, v=v_0}, \end{aligned}$$

with (u_0, v_0) a homogeneous equilibrium of the system. In this case we have

$$\begin{aligned} F_u &= 2u_0v_0 - 1 = 1 - \frac{2a}{u_0}, & F_v &= u_0^2, \\ G_u &= -2u_0v_0 = -\frac{2(u_0 - a)}{u_0}, & G_v &= -u_0^2. \end{aligned}$$

Find conditions on a such that the first three Turing conditions, eq. (18), are satisfied.

- (b) In addition, the fourth Turing condition can be shown to yield the following pair of constraints:

$$a < \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D} \right) \quad \text{or} \quad a > \frac{u_0}{2} \left(1 + \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D} \right). \quad (19)$$

Assume $D > 1$. Argue that if a satisfies the following two inequalities,

$$a < \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D} \right) \quad \text{and} \quad a > \frac{u_0(1 - u_0^2)}{2}, \quad (20)$$

then all four Turing conditions are satisfied.